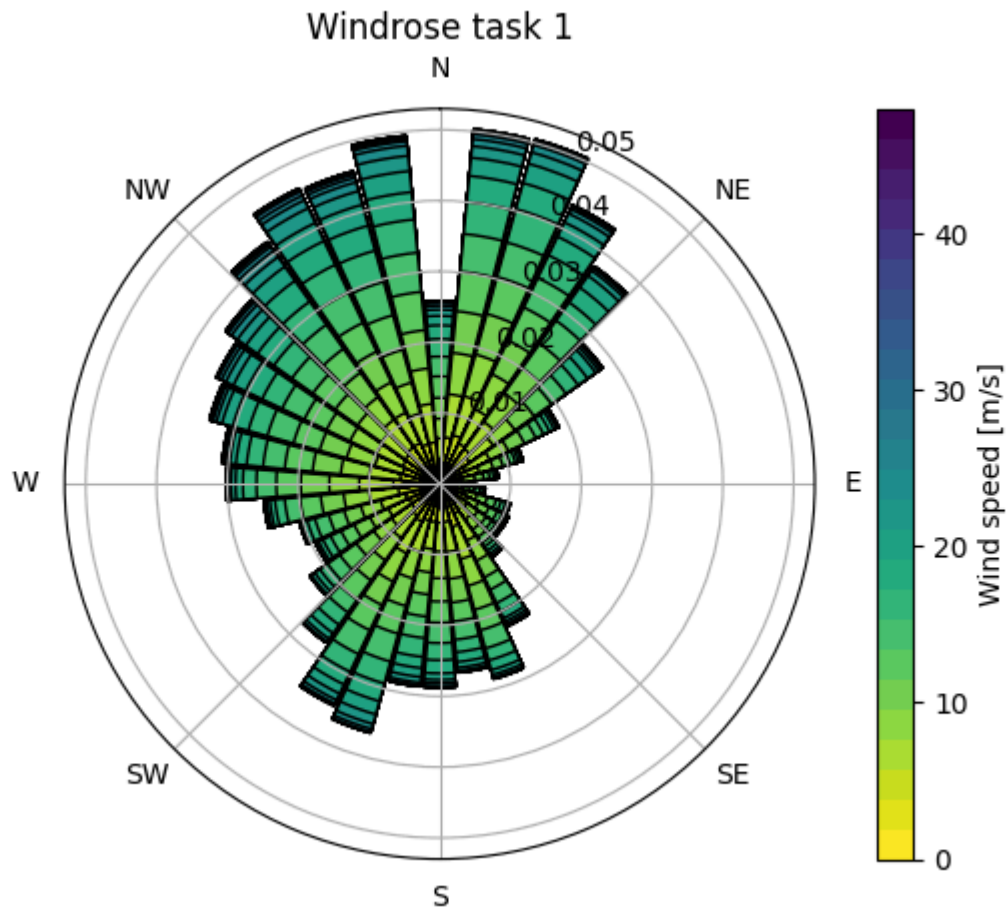


Assignment 4

Working with Floris and Landbosse.

Task 1



5° discretizations steps for wind direction.

Task 2

For task 2 I defined a second windrose object with a 1° discretization step for the wind directions and changed all requested parameters in the project_list.xlsx. One value I calculated by hand was the rated thrust.

$$T = 0.5 \cdot \rho \cdot A \cdot V^3 \cdot C_t$$

$$T = 0.5 \cdot 1.225 \text{ kg/m}^3 \cdot 13,273.23 \text{ m}^2 \cdot 96.04 \text{ m}^2/\text{s}^2 \cdot 0.8068$$

Result: The rated thrust is approximately 630 kN.

I choose a cable setup with a single cable running from the substation to the middle turbine and then a cable each to the northern and southern turbine.

AEP: 57012.82 kWh

AEP no wake: 57814.68 kWh

Cp wind farm: 0.6437

Efficiency: 0.9861 %

Task 3

When executing landbosse with the given data I received:

- project_id: foundation_validation_iea36_120
- project data: input\project_data\iea36_120_project_data.xlsx
- Annual AEP: 57,012,816 kWh
- Total BOP Cost: \$8,540,274
- Total Investment (CapEx): \$21,683,274
- Total Annual O&M (OpEx): \$1,114,405

LCOE: \$0.0475 / kWh

Task 4

LCOE Comparison:

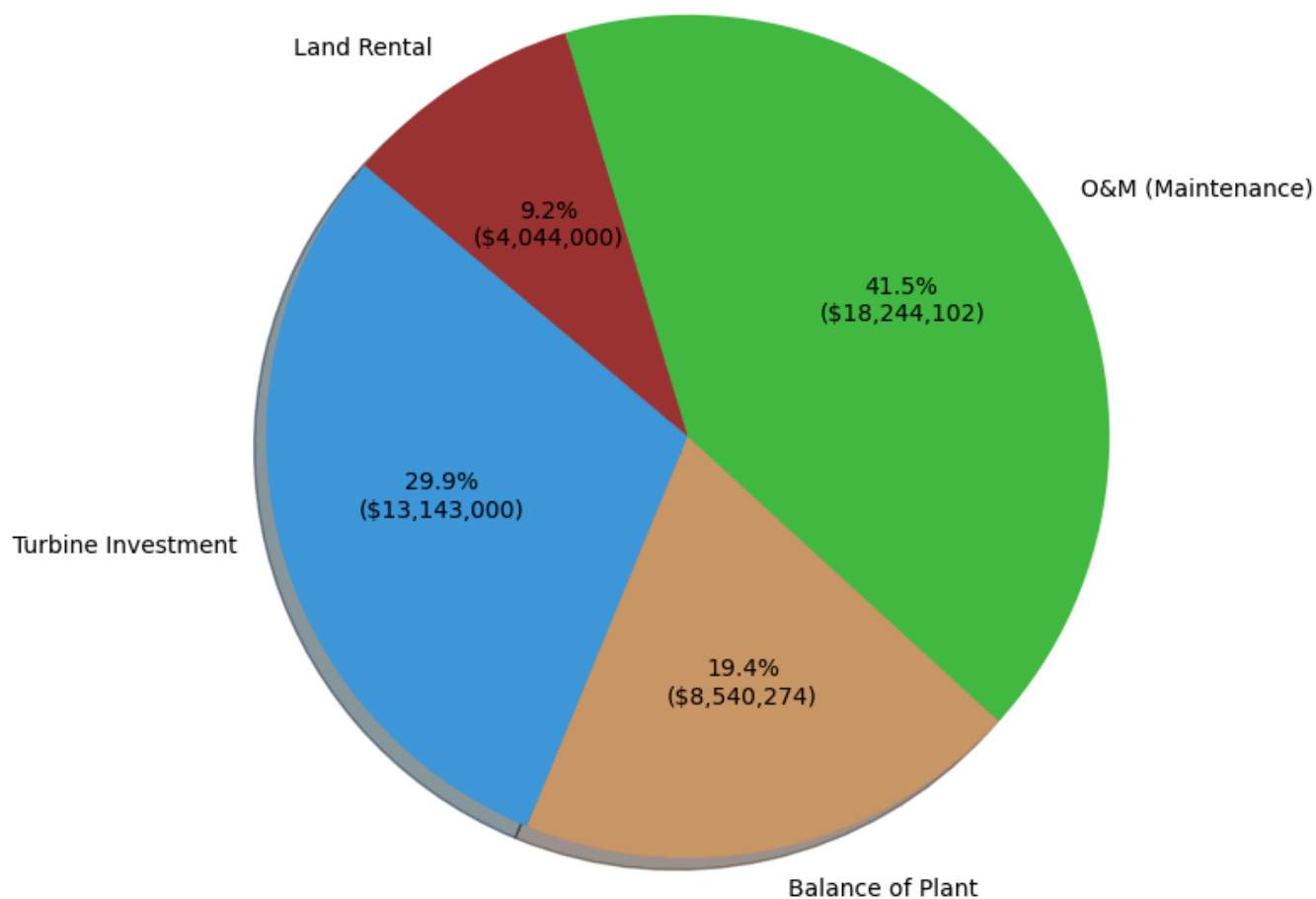
- My Calculated LCOE: \$0.0475 / kWh
- IRENA (2023 Avg): \$0.0330 / kWh
- Lazard (2024) Range: \$0.027 - \$0.075 / kWh

IRENA stands for International Renewable Energy Agency. The cited value is an industry average and stems from their published report in 2023: <https://www.irena.org/Publications/2024/Sep/Renewable-Power-Generation-Costs-in-2023> , last accessed on 12.05.2026.

Lazard is a firm active in asset management and financial advisory. They issued a report about renewable energy in 2024 defining a typical range of LCOE: https://www.lazard.com/media/xemfey0k/lazards-lcoeplus-june-2024-_vf.pdf , last accessed on 12.05.2026.

Pie chart breakdown of types of costs when building a wind farm.

Full Wind Farm Lifetime Cost Breakdown
(Total: \$43.97M)



Task 5

I re-ran floris and landbosse for 8D spacing. Because of OOP code architecture I could reuse the functions task2 and task3. With introducing a verbosity param I could silence their terminal prints for recalculations.

- AEP 5D: 57.0128 GWh | AEP 8D: 57.4287 GWh
- Relative AEP Change: +0.73%
- LCOE 5D: \$0.0475/kWh | LCOE 8D: \$0.0474/kWh
- Relative LCOE Change: -0.34%
- Observation: Increasing spacing reduced wake losses, increasing AEP.
- Observation: AEP gains from reduced wakes outweighed higher cabling costs, decreasing LCOE.

Task 6

A dropping discount rate means future revenues are discounted less heavily, while intial costs are not affected, therefore reducing the LCOE.

- Original Discount Rate: 4.0%
- New Discount Rate: 3.6%

- LCOE (Baseline): \$0.0475/kWh
- LCOE (New Rate): \$0.0465/kWh
- Relative LCOE Change: -2.07%

Task 7

This simulation is built around a small farm of only 3 turbines. In reality wind farms are often much larger (often 10+ turbines) and distribute the upfront Balance of Plant (BoP) costs—such as the substation, primary grid connection, and access roads—across a much larger energy output, significantly driving down the LCOE. Other reasons not accounted for are fluctuating energy prices and government subsidies.

Task 8

For task 8 it is necessary to calculate the cash flows for each year.

- Electricity price (averaged): \$0.0418/kWh (41.8 \$/MWh)
- Initial investment: \$21,683,274.29
- Annual revenue: \$2,383,135.72
- Annual O&M: \$1,114,405.06
- Annual net cash flow: \$1,268,730.66
- IRR (numerical): 1.55%
- PI (at 4.0% discount rate): 0.7952

Economic assessment: Project is not attractive (PI < 1).

Task 9

Calculations for increasing turbines in strict north to south direction. Note that the BoP costs rise sharply from n=4 to n=5 causing a drop in PI.

- N= 1 | LCOE=\$0.0572/kWh | PI=0.5913 | Annual Net CF=\$429,806.29 | AEP=19.27 GWh | Investment=9878041\$ | Revenue=805551\$ | BoP=5497041\$ | Turbine Capex=4381000\$
- N= 2 | LCOE=\$0.0500/kWh | PI=0.7304 | Annual Net CF=\$849,796.90 | AEP=38.16 GWh | Investment=15812954\$ | Revenue=1595200\$ | BoP=7050954\$ | Turbine Capex=8762000\$
- N= 3 | LCOE=\$0.0475/kWh | PI=0.7952 | Annual Net CF=\$1,268,730.66 | AEP=57.01 GWh | Investment=21683274\$ | Revenue=2383136\$ | BoP=8540274\$ | Turbine Capex=13143000\$
- N= 4 | LCOE=\$0.0462/kWh | PI=0.8342 | Annual Net CF=\$1,687,451.48 | AEP=75.85 GWh | Investment=27492315\$ | Revenue=3170727\$ | BoP=9968315\$ | Turbine Capex=17524000\$
- N= 5 | LCOE=\$0.2208/kWh | PI=0.1105 | Annual Net CF=\$2,106,312.17 | AEP=94.70 GWh | Investment=258997091\$ | Revenue=3958545\$ | BoP=237092091\$ | Turbine Capex=21905000\$
- N= 6 | LCOE=\$0.1906/kWh | PI=0.1300 | Annual Net CF=\$2,525,101.99 | AEP=113.55 GWh | Investment=263978728\$ | Revenue=4746247\$ | BoP=237692728\$ | Turbine Capex=26286000\$
- N= 7 | LCOE=\$0.1690/kWh | PI=0.1488 | Annual Net CF=\$2,943,923.89 | AEP=132.39 GWh | Investment=268963328\$ | Revenue=5534002\$ | BoP=238296328\$ | Turbine Capex=30667000\$
- N= 8 | LCOE=\$0.1528/kWh | PI=0.1668 | Annual Net CF=\$3,362,663.08 | AEP=151.24 GWh | Investment=273943802\$ | Revenue=6321623\$ | BoP=238895802\$ | Turbine Capex=35048000\$
- N= 9 | LCOE=\$0.1404/kWh | PI=0.1840 | Annual Net CF=\$3,781,508.52 | AEP=170.08 GWh | Investment=279359058\$ | Revenue=7109416\$ | BoP=239930058\$ | Turbine Capex=39429000\$

- N=10 | LCOE=\$0.1303/kWh | PI=0.2008 | Annual Net CF=\$4,200,325.88 | AEP=188.93 GWh | Investment=284331895\$ | Revenue=7897164\$ | BoP=240521895\$ | Turbine Capex=43810000\$
- N=11 | LCOE=\$0.1235/kWh | PI=0.2139 | Annual Net CF=\$4,619,200.47 | AEP=207.78 GWh | Investment=293471303\$ | Revenue=8685004\$ | BoP=245280303\$ | Turbine Capex=48191000\$
- N=12 | LCOE=\$0.1165/kWh | PI=0.2294 | Annual Net CF=\$5,038,035.80 | AEP=226.62 GWh | Investment=298483435\$ | Revenue=9472780\$ | BoP=245911435\$ | Turbine Capex=52572000\$
- N=13 | LCOE=\$0.1106/kWh | PI=0.2443 | Annual Net CF=\$5,456,917.83 | AEP=245.47 GWh | Investment=303564294\$ | Revenue=10260633\$ | BoP=246611294\$ | Turbine Capex=56953000\$
- N=14 | LCOE=\$0.1055/kWh | PI=0.2588 | Annual Net CF=\$5,875,765.11 | AEP=264.32 GWh | Investment=308593520\$ | Revenue=11048429\$ | BoP=247259520\$ | Turbine Capex=61334000\$
- N=15 | LCOE=\$0.1011/kWh | PI=0.2727 | Annual Net CF=\$6,294,655.15 | AEP=283.16 GWh | Investment=313683212\$ | Revenue=11836294\$ | BoP=247968212\$ | Turbine Capex=65715000\$
- N=16 | LCOE=\$0.0972/kWh | PI=0.2863 | Annual Net CF=\$6,713,515.14 | AEP=302.01 GWh | Investment=318669011\$ | Revenue=12624111\$ | BoP=248573011\$ | Turbine Capex=70096000\$
- N=17 | LCOE=\$0.0938/kWh | PI=0.2994 | Annual Net CF=\$7,132,403.87 | AEP=320.86 GWh | Investment=323744940\$ | Revenue=13411974\$ | BoP=249267940\$ | Turbine Capex=74477000\$
- N=18 | LCOE=\$0.0908/kWh | PI=0.3121 | Annual Net CF=\$7,551,278.19 | AEP=339.71 GWh | Investment=328798290\$ | Revenue=14199814\$ | BoP=249940290\$ | Turbine Capex=78858000\$
- N=19 | LCOE=\$0.0881/kWh | PI=0.3244 | Annual Net CF=\$7,970,171.88 | AEP=358.56 GWh | Investment=333860949\$ | Revenue=14987685\$ | BoP=250621949\$ | Turbine Capex=83239000\$
- N=20 | LCOE=\$0.0856/kWh | PI=0.3364 | Annual Net CF=\$8,389,047.75 | AEP=377.40 GWh | Investment=338901155\$ | Revenue=15775527\$ | BoP=251281155\$ | Turbine Capex=87620000\$
- N=21 | LCOE=\$0.0834/kWh | PI=0.3480 | Annual Net CF=\$8,807,945.64 | AEP=396.25 GWh | Investment=343990517\$ | Revenue=16563405\$ | BoP=251989517\$ | Turbine Capex=92001000\$
- N=22 | LCOE=\$0.0814/kWh | PI=0.3593 | Annual Net CF=\$9,226,829.15 | AEP=415.10 GWh | Investment=349015684\$ | Revenue=17351260\$ | BoP=252633684\$ | Turbine Capex=96382000\$
- N=23 | LCOE=\$0.0796/kWh | PI=0.3701 | Annual Net CF=\$9,645,730.47 | AEP=433.95 GWh | Investment=354157625\$ | Revenue=18139143\$ | BoP=253394625\$ | Turbine Capex=100763000\$
- N=24 | LCOE=\$0.0779/kWh | PI=0.3808 | Annual Net CF=\$10,064,609.32 | AEP=452.80 GWh | Investment=359186867\$ | Revenue=18926990\$ | BoP=254042867\$ | Turbine Capex=105144000\$
- N=25 | LCOE=\$0.0764/kWh | PI=0.3911 | Annual Net CF=\$10,483,521.34 | AEP=471.65 GWh | Investment=364301949\$ | Revenue=19714891\$ | BoP=254776949\$ | Turbine Capex=109525000\$
- N=26 | LCOE=\$0.0750/kWh | PI=0.4012 | Annual Net CF=\$10,902,402.01 | AEP=490.50 GWh | Investment=369336640\$ | Revenue=20502741\$ | BoP=255430640\$ | Turbine Capex=113906000\$
- N=27 | LCOE=\$0.0737/kWh | PI=0.4109 | Annual Net CF=\$11,321,309.90 | AEP=509.35 GWh | Investment=374484216\$ | Revenue=21290635\$ | BoP=256197216\$ | Turbine Capex=118287000\$
- N=28 | LCOE=\$0.0724/kWh | PI=0.4204 | Annual Net CF=\$11,740,194.79 | AEP=528.19 GWh | Investment=379531449\$ | Revenue=22078492\$ | BoP=256863449\$ | Turbine Capex=122668000\$
- N=29 | LCOE=\$0.0713/kWh | PI=0.4296 | Annual Net CF=\$12,159,110.60 | AEP=547.04 GWh | Investment=384665103\$ | Revenue=22866399\$ | BoP=257616103\$ | Turbine Capex=127049000\$
- N=30 | LCOE=\$0.0703/kWh | PI=0.4381 | Annual Net CF=\$12,577,997.60 | AEP=565.89 GWh | Investment=390204508\$ | Revenue=23654260\$ | BoP=258774508\$ | Turbine Capex=131430000\$
- N=31 | LCOE=\$0.0693/kWh | PI=0.4468 | Annual Net CF=\$12,996,908.78 | AEP=584.74 GWh | Investment=395367655\$ | Revenue=24442159\$ | BoP=259556655\$ | Turbine Capex=135811000\$
- N=32 | LCOE=\$0.0684/kWh | PI=0.4554 | Annual Net CF=\$13,415,804.42 | AEP=603.59 GWh | Investment=400347680\$ | Revenue=25230034\$ | BoP=260155680\$ | Turbine Capex=140192000\$

- N=33 | LCOE=\$0.0675/kWh | PI=0.4637 | Annual Net CF=\$13,834,718.25 | AEP=622.44 GWh | Investment=405464396\$ | Revenue=26017937\$ | BoP=260891396\$ | Turbine Capex=144573000\$
- N=34 | LCOE=\$0.0667/kWh | PI=0.4719 | Annual Net CF=\$14,253,610.95 | AEP=641.29 GWh | Investment=410464418\$ | Revenue=26805807\$ | BoP=261510418\$ | Turbine Capex=148954000\$
- N=35 | LCOE=\$0.0659/kWh | PI=0.4798 | Annual Net CF=\$14,672,526.94 | AEP=660.14 GWh | Investment=415597474\$ | Revenue=27593714\$ | BoP=262262474\$ | Turbine Capex=153335000\$
- N=36 | LCOE=\$0.0652/kWh | PI=0.4876 | Annual Net CF=\$15,091,424.54 | AEP=678.99 GWh | Investment=420620346\$ | Revenue=28381592\$ | BoP=262904346\$ | Turbine Capex=157716000\$
- N=37 | LCOE=\$0.0645/kWh | PI=0.4951 | Annual Net CF=\$15,510,341.76 | AEP=697.83 GWh | Investment=425772578\$ | Revenue=29169501\$ | BoP=263675578\$ | Turbine Capex=162097000\$
- N=38 | LCOE=\$0.0638/kWh | PI=0.5026 | Annual Net CF=\$15,929,233.87 | AEP=716.68 GWh | Investment=430748530\$ | Revenue=29957370\$ | BoP=264270530\$ | Turbine Capex=166478000\$
- N=39 | LCOE=\$0.0632/kWh | PI=0.5097 | Annual Net CF=\$16,348,159.82 | AEP=735.53 GWh | Investment=435899331\$ | Revenue=30745293\$ | BoP=265040331\$ | Turbine Capex=170859000\$
- N=40 | LCOE=\$0.0626/kWh | PI=0.5168 | Annual Net CF=\$16,767,052.72 | AEP=754.38 GWh | Investment=440925663\$ | Revenue=31533163\$ | BoP=265685663\$ | Turbine Capex=175240000\$

Stopped the calculation here due to computation time. Extrapolation suggests that $PI > 1$ will be reached around 115 to 125 turbines. It would be worth investigating why landbosse introduced such a large BoP cost jump.